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programs taking input from the user

1. Area of a Rectangle

Question:Write a Python program that accepts the length and width of a rectangle from the user and calculates its area.

formula: $A = \text{length} \times \text{width}$

```
In [4]: length = int(input('Enter the length of rectangle :'))
width = int(input('Enter the width of rectangle :'))
area_of_rectangle = (length * width)
area_of_rectangle
```

Out[4]: 90

2. Perimeter of a Rectangle

Question:Write a Python program that takes the length and width of a rectangle and calculates its perimeter.

formula: $P = 2(l + w)$

```
In [34]: length = int(input('enter the length of rectangle :'))
width = int(input('enter the width of rectangle :'))
Perimeter = 2*(length + width)
Perimeter
```

Out[34]: 36

3. Area of a Square

Question:Write a Python program that reads the side length of a square and finds its area.

formula : $A = \text{side} * \text{side}$

```
In [32]: side = int(input('Enter the side value of square :'))
area_of_square = side*side
area_of_square
```

Out[32]: 900

4. Perimeter of a Square

Question:Write a Python program that takes the side length of a square and calculates its perimeter.

formula : $P=4s$

```
In [35]: side = float(input("Enter side length: "))  
  
         perimeter = 4 * side  
  
         print("Perimeter of Square =", perimeter)
```

Perimeter of Square = 32.0

5. Area of a Circle

Question:Write a Python program that accepts the radius of a circle and calculates its area.

formula : $A = 3.14 * r * r$

```
In [33]: r=int(input('Enter the value for r :'))  
         area_of_circle = 3.14*r*r  
         area_of_circle
```

Out[33]: 5024.0

6. Circumference of a Circle

Question:Write a Python program that takes the radius of a circle and calculates its circumference.

formula: $C=2\pi r$

```
In [1]: radius = float(input('Enter the radius:'))  
         circumference_of_circle = 2 * 3.14 * radius  
         circumference_of_circle
```

Out[1]: 43.96

7. Area of a Triangle

Question:Write a Python program that accepts the base and height of a triangle and calculates its area.

formula: $A=1/2bh$

```
In [23]: base= eval(input('enter the base:'))
height= eval(input('enter the height:'))
Area_of_Triangle = 0.5 * base * height
Area_of_Triangle
```

Out[23]: 35.0

8. Simple Interest

Question:Write a Python program that accepts principal amount, rate of interest, and time period and calculates simple interest.

$$SI = P \times R \times T / 100$$

```
In [27]: principal_amount = float(input('Enter principal amount :'))
rate_of_interest = float(input('Enter rate of interest :'))
time_period = float(input('Enter time period :'))
simple_interest = (principal_amount * rate_of_interest * time_period) / 100
simple_interest
```

Out[27]: 41250.0

9. Total Amount Using Simple Interest

Question:Write a Python program that calculates the final amount after adding simple interest to the principal.

formula : Amount = P + SI

```
In [29]: principal_amount = float(input('Enter principal amount :'))
rate_of_interest = float(input('Enter rate of interest :'))
time_period = float(input('Enter time period :'))

simple_interest = (principal_amount * rate_of_interest * time_period) / 100
Total_amount = principal_amount + simple_interest

print("Simple interest =", simple_interest)
print("Total amount =", Total_amount )
```

Simple interest = 54000.0
Total amount = 354000.0

10. Average of Three Numbers

Question:Write a Python program that accepts three numbers and calculates their average.

formula : avg = (a + b + c) / 3

```
In [30]: num1=int(input('Enter the value for num1:'))
num2=int(input('Enter the value for num2:'))
num3=int(input('Enter the value for num3:'))
avg = (num1+num2+num3) / 3
avg
```

Out[30]: 20.0

11. Sum of First N Natural Numbers

Question:Write a Python program that accepts a positive integer N and calculates the sum of the first N natural numbers.

formula : $1+2+3+\dots+n=2n(n+1)$

```
In [1]: n = int(input("Enter a positive integer: "))
sum_n = n * (n + 1) // 2
print("Sum of first", n, "natural numbers is:", sum_n)
```

Sum of first 10 natural numbers is: 55

12. Celsius to Fahrenheit

Question:Write a Python program that converts a temperature from Celsius to Fahrenheit.

formula : $F=5C+32$

```
In [2]: celsius = float(input("Enter temperature in Celsius: "))

fahrenheit = (9/5) * celsius + 32

print("Temperature in Fahrenheit =", fahrenheit)
```

Temperature in Fahrenheit = 68.0

13. Fahrenheit to Celsius

Question:Write a Python program that converts a temperature from Fahrenheit to Celsius.

formula : $C=95(F-32)$

```
In [3]: fahrenheit = float(input("Enter temperature in Fahrenheit: "))

celsius = (fahrenheit - 32) * 5 / 9

print("Temperature in Celsius =", celsius)
```

Temperature in Celsius = 7.222222222222222

14. Area of a Parallelogram

Question:Write a Python program that accepts the base and height of a parallelogram and calculates its area.

formula : $\text{Area of Parallelogram} = \text{Base} \times \text{Height}$

```
In [4]: base = float(input("Enter the base of the parallelogram: "))
height = float(input("Enter the height of the parallelogram: "))

area = base * height

print("Area of the parallelogram =", area)
```

Area of the parallelogram = 80.0

15. Volume of a Cube

Question:Write a Python program that takes the side length of a cube and calculates its volume.

formula : $V=s^3$

```
In [5]: side = float(input("Enter the side length of the cube: "))

volume = side ** 3

print("Volume of the cube =", volume)
```

Volume of the cube = 1000.0

16. Surface Area of a Cube

Question:Write a Python program that accepts the side length of a cube and calculates its total surface area.

Formula: $A=6s^2$

```
In [6]: side = float(input("Enter the side length of the cube: "))

surface_area = 6 * (side ** 2)

print("Total Surface Area of the cube =", surface_area)
```

Total Surface Area of the cube = 216.0

17. Volume of a Cylinder

Question:Write a Python program that accepts the radius and height of a cylinder and calculates its volume.

formula : $V = \pi r^2 h$

```
In [7]: radius = float(input("Enter the radius of the cylinder: "))
height = float(input("Enter the height of the cylinder: "))

volume = 3.14 * radius ** 2 * height

print("Volume of the cylinder =", volume)
```

Volume of the cylinder = 22608.0

18. Distance Between Two Points

Question:Write a Python program that accepts coordinates $((x_1, y_1))$ and $((x_2, y_2))$ and calculates the distance between them.

```
In [1]: import math

x1 = float(input("Enter x1: "))
y1 = float(input("Enter y1: "))
x2 = float(input("Enter x2: "))
y2 = float(input("Enter y2: "))

distance = math.sqrt((x2 - x1)**2 + (y2 - y1)**2)

print("Distance between the two points =", distance)
```

Distance between the two points = 28.284271247461902

19. Percentage Calculator

Question:Write a Python program that accepts obtained marks and total marks and calculates the percentage.

Formula: $\text{Percentage} = (\text{Obtained Marks} / \text{Total Marks}) \times 100$

```
In [3]: obtained_marks = float(input("Enter obtained marks: "))
total_marks = float(input("Enter total marks: "))

percentage = (obtained_marks / total_marks) * 100

print("Percentage =", percentage, "%")
```

Percentage = 78.9 %

20. Body Mass Index (BMI)

Write a Python program that accepts a person's weight (in kilograms) and height (in meters) and calculates their Body Mass Index (BMI).

formula : $BMI = \text{Weight} \div (\text{Height} \times \text{Height})$

```
In [4]: weight = float(input("Enter your weight in kilograms: "))  
height = float(input("Enter your height in meters: "))  
  
bmi = weight / (height ** 2)  
  
print("Your BMI is =", bmi)
```

Your BMI is = 12.5

In []: